

Machines tell you when they're failing. They tell you in sound.

Applied Resonance builds hands-free acoustic diagnostics for field technicians. Smart glasses listen to the machine a technician is working on, an on-device model compares the sound against a healthy baseline, and a two-line readout on the lens flags developing faults weeks before they become breakdowns. The technician's hands never leave the job.

THE PROBLEM

Diagnosis by ear is a dying superpower

Veteran technicians hear a failing bearing across a mechanical room. That skill takes two decades to build, it was never written down, and the people who hold it are retiring faster than the trades can replace them.

Faults speak long before they fail

A mechanical fault is a geometry change: a pit in a bearing race, a shaft losing balance. Changed geometry changes forces, changed forces change vibration, and vibrating surfaces are sound. The signal appears at the first defect, weeks before function degrades. There is no silent step in that chain.

Every existing tool ignores how the work is done

A tech mid-job cannot hold a phone, watch a screen, or bolt on a sensor. Industrial monitoring instruments the machine. Nobody instruments the person standing in front of it.

WHY NOW: THREE CURVES JUST CROSSED

Audio AI fits on-device

Fault detection that needed a server now runs in real time on phone-class silicon. Public benchmarks already separate healthy machines from faulty ones.

The sensor platform shipped

Consumer smart glasses with real microphone arrays and open SDKs are on faces today. They are being used for captions; the acoustic sensor is idle.

The trades are short-handed

Skilled technicians are retiring faster than they are replaced. Y Combinator's 2026 Request for Startups names real-time AI guidance for physical work on smart glasses, with HVAC as an example vertical.

THE PRODUCT, ON THE LENS

! Bearing-like anomaly · 81%

impulse train ~88 Hz · rising · tap to log

Two glanceable lines, at most one update every two seconds. The phone holds the detail.

THE PHYSICS: FAULTS ARE PERIODIC, SO THEY ARE DIAGNOSABLE

Machines are periodic. Every internal part repeats on a schedule fixed by geometry and speed, so when a defect appears, the new sound is stamped with the rhythm of whichever part made it. The frequency itself names the culprit.

Worked example: a furnace blower bearing

Direct-drive blower at 1725 RPM gives a shaft rate of $1725 / 60 = 28.75$ Hz.

Its 6203-class bearing has 8 balls with a ball-to-pitch ratio of 0.237. The outer-race pass frequency is $\text{BPFO} = (8/2) \times 28.75 \times (1 - 0.237) = 87.8$ Hz: a tick every 11.4 ms.

Discrimination is built into the physics: imbalance thumps at exactly 28.75 Hz (1x rotation), electrical buzz sits at exactly 120 Hz (2x line frequency), the bearing ticks at 87.8 Hz.

All of it lives below 5 kHz, comfortably inside the 8 kHz bandwidth of a 16 kHz mic stream.

SYSTEM ARCHITECTURE: CAPTURE TO LENS IN SEVEN STAGES

- 1 • Capture** Four-microphone array on camera-free glasses; 16 kHz mono stream through one AudioSource interface (file, phone, or glasses: the engine never knows which).
- 2 • Windowing** 3.0 second analysis windows, 1.0 second hop, exponential smoothing on the output so the readout is stable, not flickery.
- 3 • Embedding** Each window passes through a pretrained audio network (PANNs CNN14, 2048-d embeddings), exported to ONNX and run on-device. No cloud round trip, no recordings leave the room by default.
- 4 • Scoring** Anomaly score = k-nearest-neighbor distance from the window's embedding to the active healthy baseline. Raw scores are calibrated to percentiles of the baseline's own held-out windows, so thresholds mean the same thing on every machine.
- 5 • Baselines** Three modes. Session: capture 30 seconds while the machine is presumed healthy, then watch for deviation. Saved: persist a baseline per site and machine tag, reload on the next visit (the maintenance-contract workflow). Library: cross-machine priors, the long-term roadmap as the field dataset grows.
- 6 • Alert policy** A state machine, LISTENING to SUSPECT to ALERT, with dwell (five consecutive windows above the 99th percentile), hysteresis on exit, and a cooldown. The product principle is precision over recall: it speaks rarely, and it is right when it speaks.
- 7 • Evidence** Every alert carries its why: which frequency bands deviate from baseline, envelope-spectrum peaks labeled against calculable candidates (line frequency, 1x rotation, impulse trains), and the nearest healthy windows for comparison. One 60-character evidence line reaches the lens; the claim is checkable with a wrench.

WHAT IS ALREADY MEASURED (OUR ENGINE, PUBLIC DATA)

Our detection engine is trained and evaluated on MIMII, the public industrial-sound dataset of fans and pumps recorded with factory background noise. Scorers fit on healthy clips only, per machine, matching the field reality that fault labels are scarce. Results below are AUC (higher is better, 0.5 is chance), evaluated under the DCASE2020 convention with deterministic, twice-run-identical pipelines.

Machine	Ours (kNN)	Ours (Mahalanobis)	Published baseline	Synthetic noise, 0 dB
Fan	0.749	0.773	0.658	0.811
Pump	0.893	0.900	0.729	0.879

Published baseline: the DCASE2020 Task 2 autoencoder averages. The synthetic-noise column overlays shop noise at 0 dB SNR on test clips only; the fan increase is within small-sample confidence intervals and is not treated as a finding. Current numbers are from a 400-clip development subset; the full-dataset run is in progress and will replace this table.

THE KILL TEST: DOES THE SIGNAL SURVIVE CONSUMER MICROPHONES?

Public datasets are recorded with good microphones. Glasses are not one. Before building further, we measure exactly how much detection accuracy survives a real capture chain: the test split is played through a reference speaker and re-recorded through four chains (laptop mic, phone mic, a cheap MEMS earbud as the glasses proxy, and a good-microphone control that separates speaker coloration from mic degradation). A leading sine sweep and cross-correlation re-align every clip, and the identical evaluation reruns on the degraded audio, with reference baselines kept clean and class-balanced substitution enforced so the channel cannot leak into the labels.

Pass gates, computed from our own clean margins: pump holds above 0.814 and fan above 0.699 (80 percent of the margin above chance retained). A soft fail triggers augmentation with measured impulse responses; a hard fail pivots capture to phone-records, glasses-display. The company survives every branch; only the capture path changes.

EXTERNAL EVIDENCE THE APPROACH WORKS

- Microphone-only studies classify healthy vs outer-race vs inner-race bearing faults at up to 99.26 percent accuracy with low-cost hardware; CNN approaches on acoustic spectral images report 94.6 percent across four bearing states.
- Production systems already ship: Skoda's Sound Analyser diagnoses car faults from audio at over 90 percent accuracy across roughly 245 dealerships.
- The target is the dominant failure mode: bearings account for 41 to 51 percent of motor failures across IEEE, EPRI, and ABB surveys, and 51 to 64 percent of motor failures are mechanical, which means acoustic by physics.

THE GAP: EVERYONE BUILT FOR THE MACHINE, NOBODY BUILT FOR THE TECH**Fixed industrial sensors**

Senzoro, Soundsensing. Diagnostic-grade but bolted to one machine, sold per install to factories. The roving technician never benefits.

Dealership systems

Skoda Sound Analyser, OBD devices. Real accuracy, tied to the bay and one vehicle brand.

Phone apps

Record ten seconds and hope. Consumer-grade accuracy, and the phone still occupies the hands.

Open: Applied Resonance

Diagnostic-grade, goes wherever the technician goes, hands-free by design, always on. No incumbent form factor can follow without becoming a wearable.

COVERAGE ECONOMICS AND THE MOAT

Industrial monitoring is priced per machine, so only the critical few percent of equipment ever gets a sensor. Attaching the sensor to the technician inverts the economics: one device covers every machine that person ever stands in front of, including the millions of furnaces and condensers that will never justify a bolt-on install. Each tap-to-log captures ten labeled seconds of real fault audio through wearable microphones, a dataset that exists nowhere else and that incumbents cannot collect without becoming wearables themselves. More technicians produce more data, better calls, and more technicians.

HONEST LIMITS, STATED UP FRONT

Control boards, capacitors, and thermostats fail silently; this is a second opinion alongside gauges and meters, not a replacement. Detection sits earlier than any human ear but later than fixed ultrasonic sensors. A session baseline assumes the machine sounds healthy at capture time; saved baselines close that gap on repeat visits. And a technician with twenty years of ears wins on machines they know: the wedge is the junior tech, the unfamiliar machine, and the documented evidence trail on the invoice.

ROADMAP

Now	Consumer-microphone kill test on the full capture chain; full-dataset evaluation.
Next	Engine v1: streaming inference, baseline modes, alert policy, evidence layer; live desktop demo.
Then	Apps running in both major smart-glasses simulators (Even Hub first, MentraOS second); hardware partnership outreach.
Q3 2026	Field pilots with Vancouver HVAC technicians; on-hardware mic characterization and final validation; free beta in the glasses app stores.